PRACTICAL COURSE WORKBOOK

# Financial Modelling with Excel & Python

Build a clearer, auditable financial workflow

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|---|---|
| LEVEL | Intermediate |
| GUIDED TIME | 5 hours across 12 lessons |
| PROJECT | a fictional financial model with assumptions and checks |
| SKILLS | Financial modelling, Excel, DCF, Valuation, Python, Critical evaluation |
| EDITION | Structured draft - version 0.9 |

## WHAT YOU WILL BE ABLE TO DO

- Use Financial modelling appropriately in a realistic, bounded task.
- Use Excel appropriately in a realistic, bounded task.
- Use DCF appropriately in a realistic, bounded task.
- Use Valuation appropriately in a realistic, bounded task.

## WHO THIS IS FOR

Learners studying financial technology and analytical methods.

## BEFORE YOU BEGIN

- Basic percentages and spreadsheet skills

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

# A clear route through the course

Financial Modelling with Excel & Python is a structured, practical course covering Financial modelling, Excel, DCF, Valuation, Python. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

| MODULE | LESSONS AND FOCUS |
| --- | --- |
| **01 Financial modelling: Financial foundations** | 1. Value with Financial modelling<br>2. Risk with Excel<br>3. Markets with DCF |
| **02 Excel: Tools and data** | 1. Records with Valuation<br>2. Analysis with Python<br>3. Controls with Financial modelling |
| **03 DCF: Applied decisions** | 1. Scenarios with Excel<br>2. Evaluation with DCF<br>3. Reporting with Valuation |
| **04 Valuation: Risk and governance** | 1. Security with Python<br>2. Regulation with Financial modelling<br>3. Review with Excel |

## HOW TO USE EACH LESSON PAGE

1. Read the objective and identify the decision it supports.
2. Attempt the retrieval prompt before checking the online explanation.
3. Reproduce the worked example with the stated input.
4. Test one normal case and one boundary or failure case.
5. Record the result, limitation and next action in the evidence log.

## PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

# Financial modelling: Financial foundations

Apply Financial modelling through value, risk, markets.

LESSON 01 / 18 MIN

## Value with Financial modelling

**Learning objectives**

- Explain value in the context of Financial Modelling with Excel & Python.
- Apply Financial modelling to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

value, Financial modelling, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a value result produced with Financial modelling?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 02 / 22 MIN

## Risk with Excel

**Learning objectives**

- Explain risk in the context of Financial Modelling with Excel & Python.
- Apply Excel to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

risk, Excel, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a risk result produced with Excel?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 03 / 26 MIN

## Markets with DCF

**Learning objectives**

- Explain markets in the context of Financial Modelling with Excel & Python.
- Apply DCF to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

markets, DCF, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a markets result produced with DCF?

| **Evidence to keep** | [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action |
|---|---|

MODULE REVIEW

# Turn the module into usable evidence

### 1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

### 2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

### 3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

# Excel: Tools and data

Apply Excel through records, analysis, controls.

LESSON 04 / 30 MIN

## Records with Valuation

**Learning objectives**

- Explain records in the context of Financial Modelling with Excel & Python.
- Apply Valuation to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

records, Valuation, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a records result produced with Valuation?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 05 / 18 MIN

## Analysis with Python

**Learning objectives**

- Explain analysis in the context of Financial Modelling with Excel & Python.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

analysis, Python, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a analysis result produced with Python?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 06 / 22 MIN

## Controls with Financial modelling

**Learning objectives**

- Explain controls in the context of Financial Modelling with Excel & Python.
- Apply Financial modelling to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

controls, Financial modelling, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a controls result produced with Financial modelling?

| **Evidence to keep** | [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action |
|---|---|

MODULE REVIEW

# Turn the module into usable evidence

### 1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

### 2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

### 3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

# DCF: Applied decisions

Apply DCF through scenarios, evaluation, reporting.

LESSON 07 / 26 MIN

## Scenarios with Excel

**Learning objectives**

- Explain scenarios in the context of Financial Modelling with Excel & Python.
- Apply Excel to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

scenarios, Excel, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a scenarios result produced with Excel?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 08 / 30 MIN

## Evaluation with DCF

**Learning objectives**

- Explain evaluation in the context of Financial Modelling with Excel & Python.
- Apply DCF to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

evaluation, DCF, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a evaluation result produced with DCF?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 09 / 18 MIN

## Reporting with Valuation

**Learning objectives**

- Explain reporting in the context of Financial Modelling with Excel & Python.
- Apply Valuation to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

reporting, Valuation, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a reporting result produced with Valuation?

| Evidence to keep | [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action |
|---|---|

MODULE REVIEW

# Turn the module into usable evidence

### 1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

### 2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

### 3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

# Valuation: Risk and governance

Apply Valuation through security, regulation, review.

LESSON 10 / 22 MIN

## Security with Python

**Learning objectives**

- Explain security in the context of Financial Modelling with Excel & Python.
- Apply Python to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

security, Python, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a security result produced with Python?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 11 / 26 MIN

## Regulation with Financial modelling

**Learning objectives**

- Explain regulation in the context of Financial Modelling with Excel & Python.
- Apply Financial modelling to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

regulation, Financial modelling, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a regulation result produced with Financial modelling?

**Evidence to keep**

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 12 / 30 MIN

## Review with Excel

**Learning objectives**

- Explain review in the context of Financial Modelling with Excel & Python.
- Apply Excel to a bounded practical task.
- Evaluate the result using explicit quality criteria.

**Key terms**

review, Excel, finance

**Retrieval prompt**

In Financial Modelling with Excel & Python, which evidence best supports a review result produced with Excel?

| | |
|---|---|
| **Evidence to keep** | [ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action |

MODULE REVIEW

# Turn the module into usable evidence

| 1. RETRIEVE |
|---|
| Without checking the lessons, explain the module's three most important ideas in your own words. |
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| 2. APPLY |
|---|
| Describe the example or file you produced, the input used, and the result you observed. |
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| 3. REVIEW |
|---|
| Record one uncertainty, one source to recheck, and the next deliberate practice action. |
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CAPSTONE PROJECT

# Produce a reviewable Financial Modelling with Excel & Python project using Financial modelling, Excel, DCF.

| | |
|---|---|
| DELIVERABLE | A working Financial Modelling with Excel & Python artefact plus an evidence-based self-review. |
| TOOLS | A suitable Financial modelling environment, A plain-text decision log, Test data or realistic sample material |

## Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using Financial modelling.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

## Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This course is general education, not financial, investment, tax or legal advice.

PROJECT EVIDENCE

# Document decisions another person can review

| 1. PURPOSE AND USER |
| --- |
| State the intended user, need, expected outcome and constraints. |
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| 2. INPUTS AND ASSUMPTIONS |
| --- |
| List source files, data, versions, permissions and assumptions. |
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PROJECT EVIDENCE

# Tests, limitations and revision

### 3. NORMAL CASE

Expected result, observed result and evidence location.

### 4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

### 5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

# Make the work credible and repeatable

## Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

| FINAL DECISION |
| --- |
| Is the project ready to share? State the evidence, remaining risk, owner and next review date. |
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## Authoritative references

### National Payments System

Central Bank of Kenya - reviewed 2026-08-21

https://www.centralbank.go.ke/national-payments-system/

### Financial education

OECD - reviewed 2026-08-21

https://www.oecd.org/en/topics/financial-education.html

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SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Financial modelling scope.